Z

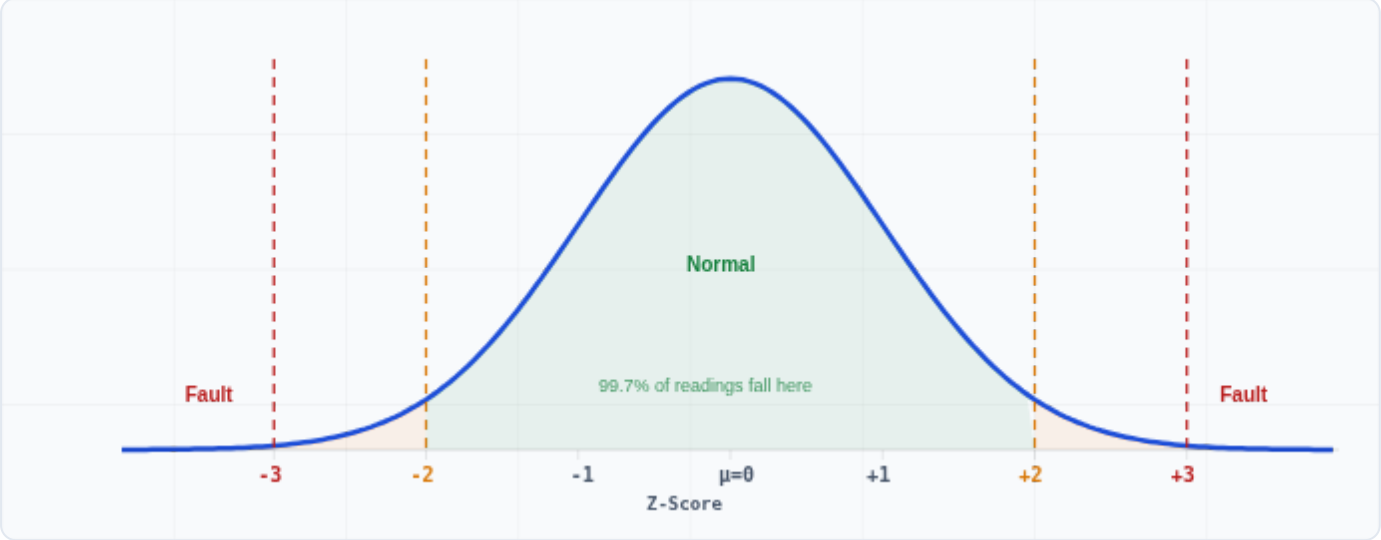
The Z-Score

The "Truth Meter" — how we give signals a universal language

CORE CONCEPT · STATISTICAL INTELLIGENCE

THE FORMULA $Z = (x - \mu) / \sigma$

x = current reading · μ = mean of healthy data · σ = standard deviation



BELL CURVE (NORMAL DISTRIBUTION) · Z-SCORE ZONES FROM −3 TO +3 · FAULT DETECTION THRESHOLD ILLUSTRATED

Every sensor reading your motor produces — RMS, Kurtosis, ZCR — is just a raw number. Alone, it tells you almost nothing. **RMS = 0.42** could mean the motor is healthy, or it could be drowned by crowd noise. You have no reference.

The Z-Score solves this by answering one simple question: **"How far is this reading from what's normal?"** It converts any raw number into a universal scale where **Z = 0 means perfectly normal** and **Z = ±3 means something is very wrong**. Suddenly, RMS, frequency, and Kurtosis all speak the same language.

Z = 0
PERFECT NORMAL
The reading equals the historical mean exactly. Motor is running as expected.

Z = ±2
SUSPICIOUS
Slightly outside normal. Log it. Begin monitoring for trend.

Z = ±3
FAULT ALERT
Only 0.3% of normal readings reach here. Trigger the Industrial Doctor alert.

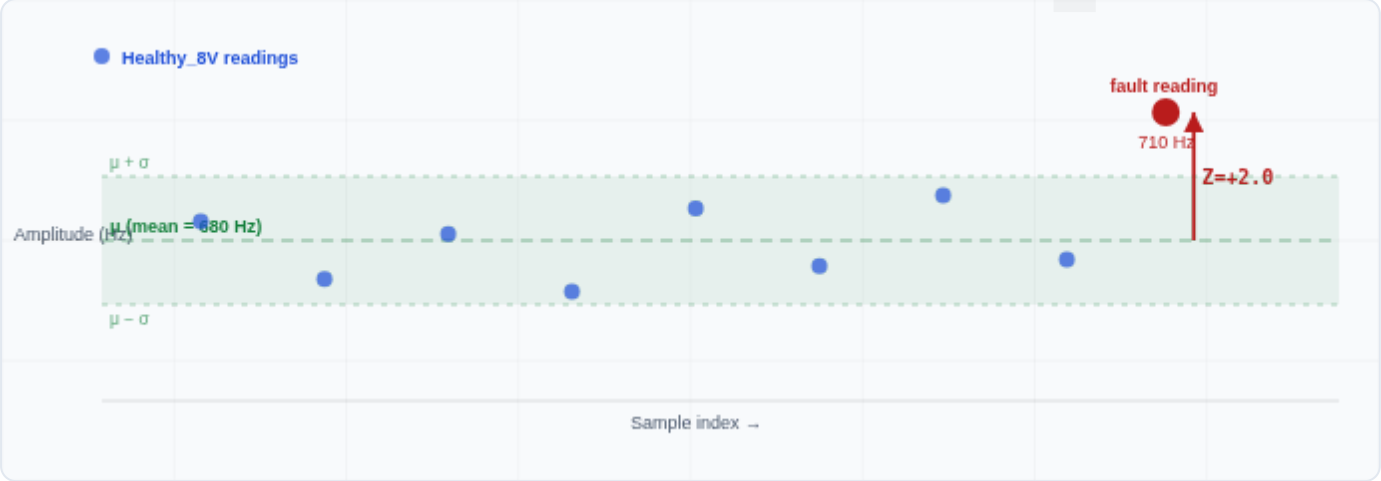
Z > ±4
CRITICAL
Statistically near-impossible in healthy operation. Immediate stop recommended.

μσ

The Formula — Unpacked

What each variable means and how they behave in a real motor scenario

FORMULA BREAKDOWN · WORKED EXAMPLE



REAL DATA STRIP — HEALTHY 8V BASELINE (BLUE BAND = M ± Σ) · FAULT READING SHOWN AS RED DOT · Z-SCORE COMPUTED LIVE

X — CURRENT READING

The raw value the sensor just measured. In our project this is one feature — e.g., Peak Frequency = **710 Hz**. This is what just happened.

M — THE MEAN (YOUR BASELINE)

The average of all **Healthy_8V** readings you collected. This is what "normal" looks like for your specific motor at this voltage. μ = **680 Hz**.

Σ — STD DEVIATION (STABILITY)

How much the healthy readings naturally vary. A tight motor with consistent behaviour has a **small σ**. A noisy environment has a large σ. Here σ = **15 Hz**.

Worked example using Peak Frequency for Healthy_8V data:

1. The MEMS microphone captures a vibration window. After FFT, the dominant peak is at **710 Hz**. This is your raw reading **x = 710**.

2. From your **Healthy_8V dataset**, you computed that the mean peak frequency μ = **680 Hz** and the standard deviation σ = **15 Hz**.

3. Apply the formula: **Z = (710 − 680) / 15 = 30 / 15 = +2.0**. The reading is 2 standard deviations above the healthy average — suspicious but not yet an alert.

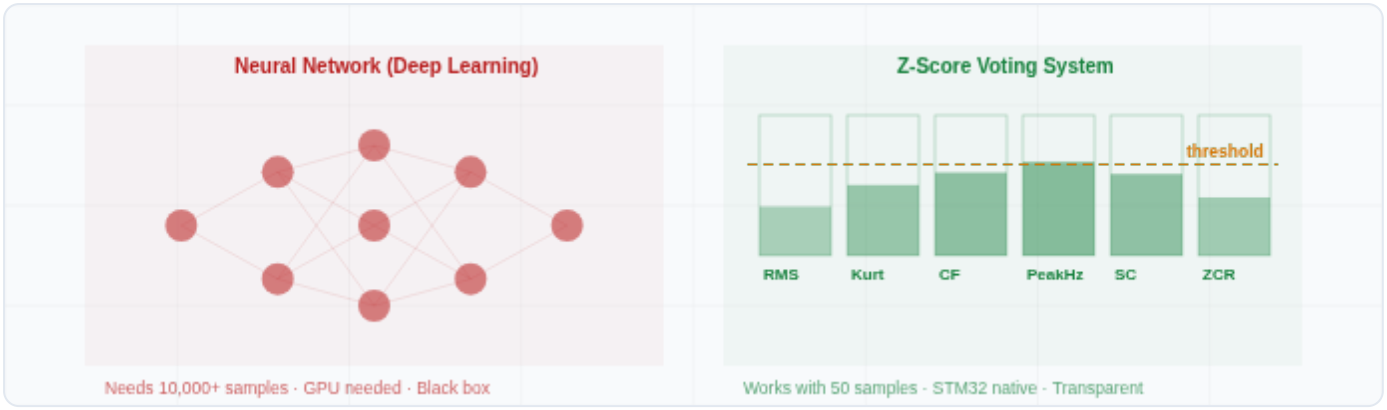
4. The voting system logs **Z = +2.0** for Peak Frequency. If the next window gives Z = +2.6, and RMS also shows Z = +2.2, the combined vote crosses the threshold → **Industrial Doctor Alert fired.**

VS

Standard AI vs. Z-Score Strategy

Why we chose statistical intelligence over deep learning for this project

ARCHITECTURE DECISION · WHY Z-SCORE



CONCEPTUAL COMPLEXITY VS DEPLOYMENT COST — NEURAL NETWORK (LEFT) VS Z-SCORE VOTING (RIGHT)

CRITERION	STANDARD AI (NEURAL NET)	Z-SCORE VOTING (OUR APPROACH)
Data required	**Thousands of labelled fault samples — we simply don't have these yet**	**Dozens of clean healthy samples. We already have these.**
Hardware cost	**Needs GPU or powerful CPU for inference. Won't run on STM32.**	**Just arithmetic. Runs in microseconds on any microcontroller.**
Transparency	**"Black Box" — you cannot see why it classified a fault.**	**Every vote is traceable: "RMS Z=+3.1 triggered the alert."**
Calibration	**Requires full retraining when motor model changes.**	**Just recalculate μ and σ from new healthy data — minutes of work.**
Real-time speed	**Inference latency 10–500ms depending on model size.**	**Sub-millisecond. Decisions made faster than data arrives.**
Explainability	**Cannot tell an engineer which physical component is failing.**	**High Kurtosis Z-score → bearing. High SC Z-score → friction.**

THE REAL PROBLEM WITH NEURAL NETWORKS HERE

A neural network learns "fault" by seeing thousands of fault examples. But your motor hasn't failed yet — that's the point. You are trying to **predict the first failure**, not classify from existing ones. Z-Score detects the drift towards failure using only normal data as a reference.

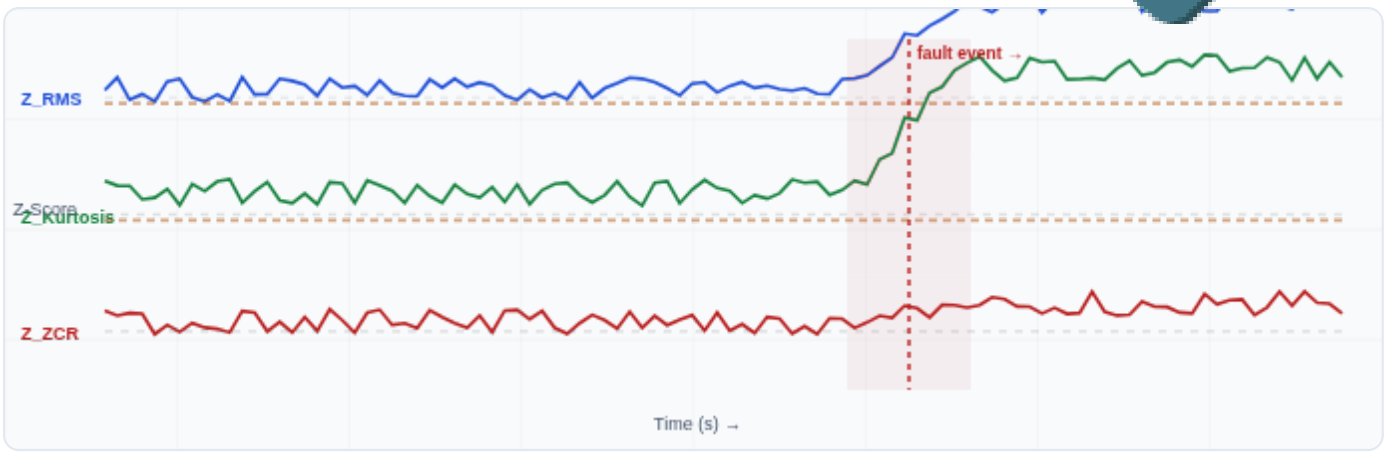
THE Z-SCORE ADVANTAGE FOR INDUSTRIAL DOCTOR

Our system only needs to know what **healthy sounds like**. Anything that deviates significantly from that profile — regardless of what it is — gets flagged. This is called **anomaly detection**, and it is the right paradigm for predictive maintenance.

Z-Score in Sound Signal Analysis

Five concrete ways Z-Score transforms raw audio features into meaningful intelligence

SOUND ANALYSIS · PRACTICAL APPLICATIONS



THREE SIMULTANEOUS Z-SCORE TRACKS (RMS, KURTOSIS, ZCR) OVER TIME · FAULT EVENT VISIBLE AS SPIKE AT T=6S

1. Dynamic Thresholding

Instead of hardcoding "alert if Volume > 500", Z-score says **"alert if Volume is 3× higher than its usual background level."** Works in a quiet lab and a noisy factory floor — same code, same threshold.

2. Ambient Noise Veto

Crowd noise spikes **RMS Z-score** but leaves **Peak Frequency Z-score flat**. The voting system sees this mismatch and vetoes the false alarm — no false positive from a noisy factory break room.

3. Sensor Fusion

Microphone gives Volume in dB. Accelerometer gives vibration in m/s². Completely different units — impossible to combine. **Convert both to Z-scores** and they speak the same language: deviation from normal.

4. Trend Detection

A single spike means nothing. But if Kurtosis Z-score **slowly climbs from +1.0 to +2.8 over 3 days**, the bearing is degrading. Z-scores make gradual trends visible before they become failures.

5. Outlier Immunity

A random click or tap creates one huge spike. Z-score of that single sample is enormous — but averaging Z-scores over a 1-second window **absorbs isolated spikes** and keeps the system calm.

6. Feature Normalisation

RMS is ~0.4, Kurtosis is ~3.2, ZCR is ~0.17. All different scales — you can't average them. **Z-scores normalise all six features to the same −3 to +3 range**, making a weighted vote possible and fair.

The Z-Score Voting System

How all six features combine into a single fault decision for Industrial Doctor

SYSTEM ARCHITECTURE · DECISION ENGINE



VOTING BAR CHART — EACH FEATURE CASTS A Z-SCORE VOTE · THRESHOLD LINE AT Z=2.5 · MAJORITY VOTE TRIGGERS ALERT

No single feature is reliable alone — RMS is fooled by noise, Kurtosis can spike randomly. The **Voting System** asks all six features simultaneously: **"Are you anomalous right now?"** Each one casts a vote weighted by its Z-score. Only when the **combined weighted vote exceeds the threshold** does the alert fire — this is what eliminates false positives.

① **Collect baselines.** Run the motor healthy at 6V and 8V. Record mean (μ) and std deviation (σ) for all 6 features. Store in flash memory on the STM32. This is done once.

② **Per-window scoring.** Every 512ms, extract the 6 features from the microphone buffer. Compute Z = (x − μ) / σ for each. You now have 6 Z-scores: Z_RMS, Z_Kurt, Z_CF, Z_PeakHz, Z_SC, Z_ZCR.

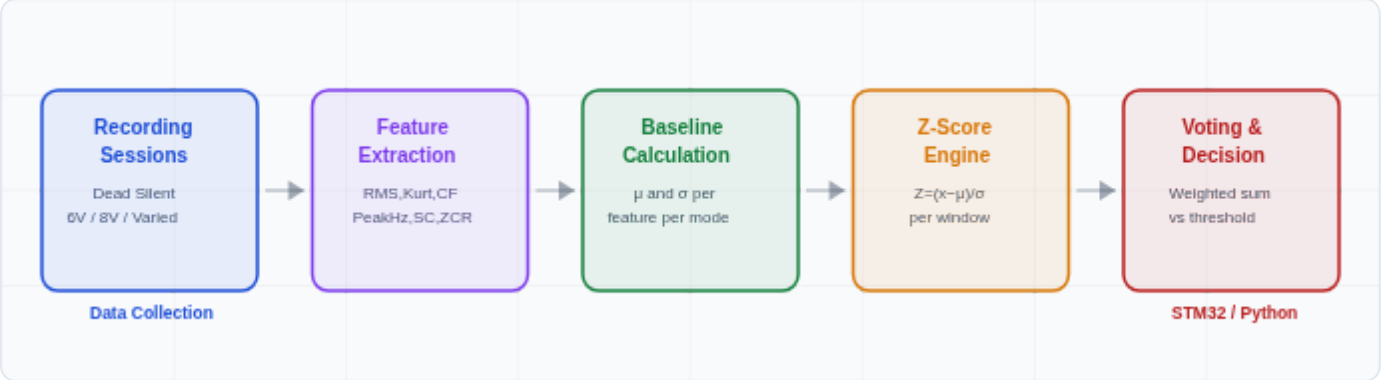
③ **Weighted vote.** Sum the Z-scores with weights based on fault sensitivity: **Kurtosis × 2.0, Spectral Centroid × 1.8, Crest Factor × 1.5**, others × 1.0. Divide by total weights to get a normalised fault score.

④ **Threshold decision.** If the weighted fault score exceeds **+2.5** for three consecutive windows, the Industrial Doctor fires the alert, logs which features voted highest, and reports the probable fault type.

Dataset Strategy & Complete Summary

How each recording session feeds the Z-score engine — and what comes next

DATA PIPELINE · PROJECT ROADMAP



DATA PIPELINE — FROM RAW RECORDING SESSIONS TO Z-SCORE BASELINES TO LIVE FAULT DETECTION

DEAD SILENT SESSION → FLOOR BASELINE

Provides the **ambient noise floor**: μ and σ of all features when nothing is running. Used to detect "is the motor even on?" and to separate environmental noise from motor signals.

HEALTHY 6V / 8V → OPERATING BASELINES

The most important datasets. Provide the **μ and σ for normal operation** at each voltage. All Z-scores are computed relative to these. The better these baselines, the more precise the fault detection.

HEALTHY VARIED → ROBUSTNESS BASELINE

Captures natural variation during real operation: slight load changes, temperature drift, motor warm-up. **Widens σ appropriately** so the system doesn't false-alarm on normal variation in real deployment.

CROWD / AMBIENT → VETO CALIBRATION

Teaches the system what **environmental noise looks like in Z-score space**. Specifically: high Z_RMS but flat Z_PeakHz and chaotic Z_ZCR → ambient label, suppress alert.

The complete picture: Your Z-Score engine is not a generic AI — it is a precise, calibrated instrument built specifically for your motor, your environment, and your MEMS hardware. Every recording session you complete makes the μ and σ values more accurate, which directly tightens the Z-score boundaries and reduces both false positives and missed faults. **More data = sharper detection, no retraining required.**